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EVOLUTION AND INTERACTION OF LARGE INTERPLANETARY STREAMS

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Abstract. The paper presents a computer simulation for the evolution and interaction of large interplanetary streams based on multi-spacecraft observations and an unsteady, one-dimensional MHD model. We studied two events each observed by two or more spacecraft separated by a distance of the order of 10 AU. The first simulation is based on the plasma and magnetic field observations made by two radially-aligned spacecraft: first by IMP-8 in November 1977 at 1 AU, and later by Pioneer-10 in January 1978 at 15 AU. The second simulation is based on an event observed first by Helios-1 in May 1980 near 0.6 AU and later by Voyager-1 in June 1980 at 8.1 AU. These examples show that the dynamical evolution of large-scale solar wind structures is dominated by the shock process, including the formation, collision and merging of shocks. Formation of shocks continues to take place even outside 5 AU. Collision and merging of shocks irreversibly restructure the solar wind. The interaction of shocks with stream structures also causes a drastic decrease in the amplitude of the solar wind speed variation with increasing heliocentric distance, and as a result of interactions there is a large variation of shock-strengths and shock-speeds. In the outer heliosphere, the large-scale solar wind and magnetic field evolve into a much simpler structure and MHD shocks are present as a principal component of the solar wind. The simulation results shed new light on interpretation for the interaction and evolution of large interplanetary streams. Observations were made only along a few limited trajectories, but simulation results can supplement these by providing the detailed evolution process for large-scale solar wind structures in the vast region not directly